
Hayden Austin

Software Engineer | Machine Learning Systems

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Software engineer with independent work in recurrent models, GPU programming and numerical validation. Implementations span PyTorch, JAX/Flax, CUDA and Triton, supported by reference tests, controlled benchmarks and reproducible research tooling.

RESEARCH & MACHINE LEARNING PROJECTS

Independent ML systems & numerical research | [Code](#)

- Implemented recurrent state-space models in PyTorch and JAX/Flax, with full-sequence, chunked and token-by-token execution.
- Developed parallel scans and C++/CUDA and Triton kernels with custom PyTorch autograd forward/backward integration.
- Built reference tests for outputs, gradients and streaming equivalence, including near-zero scale factors and CUDA-stream edge cases.
- Built reproducible benchmarks with pinned data and matched comparison settings; verified research artifacts with SHA-256 manifests.
- Used SymPy/FLINT for exact-arithmetic checks, alongside independent numerical validation of structured operators.

Animal-image classification | [Notebook](#)

- Built a TensorFlow/Keras CNN with a stratified train/validation split and a tf.data pipeline for preprocessing, batching and prefetching.
- Evaluated validation-set predictions with a confusion matrix; saved and reloaded the trained model for an inference demonstration.

EXPERIENCE

Freelance Developer

2023 - 2025

- Developed web applications for commerce and professional services, covering product catalogues, authentication, content management and deployment.

Game Server Developer & Administrator

2019 - 2022

- Built event-driven Java plugins with persistent item metadata and server API integrations; maintained Linux servers and investigated performance issues.

TECHNICAL SKILLS

Models & training: Python, PyTorch, JAX/Flax, TensorFlow/Keras

GPU & numerical: C++, CUDA, Triton, NumPy, SymPy, FLINT

Validation: Autograd, reference tests, numerical checks, benchmark tooling

Engineering: Git, Linux, automated testing; React, TypeScript and Firebase

EDUCATION

Belgium Campus ITversity

Studies toward a Bachelor of Computing, specialising in Software Engineering.